MATHEMATICS OF CRYPTOGRAPHY · MODULE 01

Euclid and Modular Arithmetic

Divisibility and the Euclidean Algorithm, then Modular Arithmetic and Congruences

2-hour session

2 lessons + 24 quiz questions

```
gcd(252, 105) = 21
7 × 8 ≡ 1 (mod 11)
7^256 mod 13 = 9
```

Dr.-Ing. Chukwuka Victor Obionwu

PLAN · 2 HOURS

Session timeline

Part A · Lesson 1

Time	Segment	Min
0:00	Welcome & the RSA puzzle	5
0:05	Divisibility & division	8
0:13	GCD & Euclid	10
0:23	Quiz 1	7
0:30	Bézout & Extended Euclid	12
0:42	Quiz 2	8
0:50	Break	5

Part B · Lesson 2

Time	Segment	Min
0:55	Congruences & the ring	13
1:08	Quiz 3	8
1:16	Inverses & calculator	10
1:26	Square-and-multiply	14
1:40	Quiz 4	10
1:50	Recap & exit ticket	10

OBJECTIVES

By the end of today you can…

1. Use divisibility and the division algorithm
2. Compute a gcd with Euclid and explain why it is fast
3. Find Bézout coefficients and modular inverses
4. Work in Z_m and compute g^e mod m by square-and-multiply

THE PUZZLE · 0:02

Generating an RSA key needs a number d that undoes multiplication by 17.

$$17 \times d \equiv 1 \pmod{3120}$$

Trying all 3,120 candidates is hopeless at real key sizes. We need a method.

WARM-UP · 2 MIN

Does a solution d always exist? What would have to be true about 17 and 3120?

a divides b

a | b if there is an integer c such that $b = a \cdot c$

7 | 91

Because 91 = 7 · 13.

6 does not divide 20

No integer c gives 6 · c = 20. (6 · 3 = 18 and 6 · 4 = 24.)

5 | 0

Because 0 = 5 · 0. Every integer divides 0.

Divisibility is the foundational concept of number theory. Everything today is built on it.

Quotient and remainder, uniquely

For integers a and b > 0 there are unique q and r with $a = b \cdot q + r,\ 0 \le r < b$

a	b	q	r	Check
47	5	9	2	5·9 + 2
100	7	14	2	7·14 + 2
−17	5	−4	3	5·(−4) + 3

THINK · 1 MIN

For a = −17 and b = 5, why is q = −3 with r = −2 not allowed?

Hint: look at the condition on r.

Greatest common divisor

gcd(a, b) is the largest positive integer d that divides both a and b.

If gcd(a, b) = 1, then a and b are coprime (relatively prime).

Pair	Common divisors	gcd	Coprime?
(12, 18)	1, 2, 3, 6	6	No
(35, 15)	1, 5	5	No
(8, 15)	1	1	Yes

One identity does all the work

$$\gcd(a, b) = \gcd(b, a \bmod b)$$

WHY IT WORKS

Write a = b·q + r. A number that divides both a and b also divides r = a − b·q. A number that divides b and r also divides a. So both pairs have the same common divisors.

WHY IT IS FAST

No factoring needed, which is computationally hard for large numbers. The algorithm ends after O(log(min(a, b))) steps, by Lamé's Theorem.

Repeat until the remainder is 0. The last nonzero remainder is the gcd.

WORKED EXAMPLE · 0:20

gcd(252, 105) by hand

Step 1 $252 = 2 \times 105 + \mathbf{42}$

Step 2 $105 = 2 \times 42 + \mathbf{21}$

Step 3 $42 = 2 \times 21 + \mathbf{0}$

gcd(252, 105) = 21

PAGE CALCULATOR EXAMPLE

$13 = 3 \times 4 + 1$

$4 = 4 \times 1 + 0$

So gcd(13, 4) = 1.

YOUR TURN · 2 MIN

Compute gcd(91, 35) with the same steps.

QUIZ 1 · 0:23

Questions 1–3

QUESTION 1

Which statement defines a | b (a divides b)?

- **A** a = b · c for some integer c
- **B** a + c = b for some integer c
- **C** b = a · c for some integer c
- **D** a · b = c for some integer c

QUESTION 2

Dividing 47 by 5 gives which quotient and remainder?

- **A** q = 8, r = 7
- **B** q = 9, r = 2
- **C** q = 10, r = −3
- **D** q = 9, r = 5

QUESTION 3

What is gcd(84, 36)?

- **A** 6
- **B** 9
- **C** 18
- **D** 12

Vote by show of hands, then reveal on the next slide.

QUIZ 1 · ANSWERS

Answers 1–3

QUESTION 1

Which statement defines a | b (a divides b)?

Answer: C

b = a · c for some integer c

a | b means b is an integer multiple of a: b = a · c.

QUESTION 2

Dividing 47 by 5 gives which quotient and remainder?

Answer: B

q = 9, r = 2

47 = 5 · 9 + 2, and 0 ≤ 2 < 5.

QUESTION 3

What is gcd(84, 36)?

Answer: D

12

84 = 2·36 + 12, then 36 = 3·12 + 0. The last nonzero remainder is 12.

QUIZ 1 · 0:27

Questions 4–6

QUESTION 4

True or false: gcd(a, b) = gcd(b, a mod b).

- **A** True
- **B** False

QUESTION 5

Which pair of numbers is coprime?

- **A** (35, 12)
- **B** (12, 18)
- **C** (21, 14)
- **D** (30, 45)

QUESTION 6

How many steps does Euclid take, by Lamé's Theorem?

- **A** O(min(a, b))
- **B** O(a · b)
- **C** O(2^n)
- **D** O(log(min(a, b)))

Vote by show of hands, then reveal on the next slide.

QUIZ 1 · ANSWERS

Answers 4–6

QUESTION 4

True or false: gcd(a, b) = gcd(b, a mod b).

Answer: A

True

True. This identity is the heart of the Euclidean algorithm.

QUESTION 5

Which pair of numbers is coprime?

Answer: A

(35, 12)

gcd(35, 12) = 1. The others have gcd 6, 7 and 15.

QUESTION 6

How many steps does Euclid take, by Lamé's Theorem?

Answer: D

O(log(min(a, b)))

The number of steps grows logarithmically in the smaller input.

LESSON 1 · BÉZOUT · 0:30

The gcd is a combination of a and b

There exist integers x and y such that $a \cdot x + b \cdot y = \gcd(a, b)$

EXAMPLE

$35 \cdot 1 + 15 \cdot (-2) = 5$

Here gcd(35, 15) = 5, with x = 1 and y = −2.

WHY CRYPTOGRAPHERS CARE

If gcd(a, m) = 1 then $a \cdot x + m \cdot y = 1$. Reduce mod m and $a \cdot x \equiv 1$. So x is the modular inverse of a.

PREDICT · 1 MIN

Is the pair (x, y) unique? Try x = 1 + 3 = 4 for 35x + 15y = 5. What must y be?

WORKED EXAMPLE · 0:34

Back-substitute to get x and y

FORWARD (EUCLID)

$$252 = 2 \cdot 105 + 42$$

$$105 = 2 \cdot 42 + 21$$

BACKWARD

$$21 = 105 - 2 \cdot 42$$

$$42 = 252 - 2 \cdot 105$$

$$21 = 105 - 2 \cdot (252 - 2 \cdot 105)$$

$$21 = 5 \cdot 105 - 2 \cdot 252$$

gcd = 21

x = −2, y = 5

for a = 252, b = 105

CHECK · 1 MIN

Verify: $252 \cdot (-2) + 105 \cdot 5 = ?$

CODE · 0:38

Extended Euclid in Python

```
def extended_gcd(a, b):
    if b == 0:
        return (a, 1, 0)
    g, x1, y1 = extended_gcd(b, a % b)
    x = y1
    y = x1 - (a // b) * y1
    return (g, x, y)
```

TRY IT

`extended_gcd(252, 105)`

`(21, −2, 5)`

BASE CASE

gcd(a, 0) = a, because a = a·1 + 0·0.

BACK TO THE PUZZLE · 0:40

Solving 17 · d ≡ 1 (mod 3120)

THE METHOD

1. Run extended_gcd(17, 3120).
2. Confirm the gcd is 1.
3. Take x % 3120 as the inverse.

```
extended_gcd(17, 3120)
(1, −367, 2)
x = −367
−367 % 3120 = 2753
```

17 · 2753 = 46801 = 15 · 3120 + 1

Warm-up answered: d exists because gcd(17, 3120) = 1, and d = 2753.

QUIZ 2 · 0:42

Questions 1–3

QUESTION 1

Bézout’s identity says there exist integers x and y with…

- **A** $a \cdot x + b \cdot y = a + b$
- **B** $a \cdot x - b \cdot y = 0$
- **C** $a \cdot x + b \cdot y = \gcd(a, b)$
- **D** $a \cdot x \cdot b \cdot y = 1$

QUESTION 2

Which (x, y) satisfies 35x + 15y = 5?

- **A** (−1, 2)
- **B** (1, −2)
- **C** (2, −1)
- **D** (3, −1)

QUESTION 3

What is the inverse of 7 modulo 11?

- **A** 8
- **B** 4
- **C** 5
- **D** 3

Vote by show of hands, then reveal on the next slide.

QUIZ 2 · ANSWERS

Answers 1–3

QUESTION 1

Bézout's identity says there exist integers x and y with…

Answer: C

a·x + b·y = gcd(a, b)

The gcd is always a combination a·x + b·y of a and b.

QUESTION 2

Which (x, y) satisfies 35x + 15y = 5?

Answer: B

(1, −2)

35·1 + 15·(−2) = 35 − 30 = 5.

QUESTION 3

What is the inverse of 7 modulo 11?

Answer: A

8

7 · 8 = 56 = 5·11 + 1, so 7 · 8 ≡ 1 (mod 11).

QUIZ 2 · 0:46

Questions 4–6

QUESTION 4

True or false: a has an inverse modulo m exactly when gcd(a, m) = 1.

- **A** True
- **B** False

QUESTION 5

In extended_gcd(a, b), what does the base case b = 0 return?

- **A** (0, 1, 0)
- **B** (a, 1, 0)
- **C** (a, 0, 1)
- **D** (1, 0, 0)

QUESTION 6

In the RSA example, inv = x % 3120 with x = −367 gives…

- **A** −367
- **B** 367
- **C** 2753
- **D** 3120

Vote by show of hands, then reveal on the next slide.

QUIZ 2 · ANSWERS

Answers 4–6

QUESTION 4

True or false: a has an inverse modulo m exactly when gcd(a, m) = 1.

Answer: A

True

True. If gcd(a, m) = 1, Bézout gives a·x + m·y = 1, so a·x ≡ 1.

QUESTION 5

In extended_gcd(a, b), what does the base case b = 0 return?

Answer: B

(a, 1, 0)

gcd(a, 0) = a, and a·1 + 0·0 = a, so (g, x, y) = (a, 1, 0).

QUESTION 6

In the RSA example, inv = x % 3120 with x = −367 gives…

Answer: C

2753

−367 + 3120 = 2753, so Python’s % maps x into 0..3119.

BREAK · 0:50–0:55

Five minutes. Back at 0:55.

WHILE YOU STRETCH

Is 23 the inverse of 7 modulo 40? Decide with one multiplication.

Next: congruences, the ring Z_m, and fast modular exponentiation.

Same remainder, same class

a ≡ b (mod m) means m divides (a − b)

23 ≡ 3 (mod 5)

23 − 3 = 20 and 5 | 20.

17 ≡ 5 (mod 12)

17 − 5 = 12 and 12 | 12.

−3 ≡ 2 (mod 5)

−3 − 2 = −5 and 5 | −5.

ASK · 1 MIN

Is 19 ≡ 4 (mod 6)? Test the difference.

Congruence sorts the integers into m classes

$$[a] = \{\, a + k \cdot m : k \text{ is an integer} \,\}$$

Class	Members (mod 5)
[0]	…, −10, −5, 0, 5, 10, 15, …
[1]	…, −9, −4, 1, 6, 11, 16, …
[2]	…, −8, −3, 2, 7, 12, 17, …
[3]	…, −7, −2, 3, 8, 13, 18, …
[4]	…, −6, −1, 4, 9, 14, 19, …

Every integer lies in exactly one class. Classes never overlap.

Modulus m gives exactly m classes: [0], [1], …, [m − 1].

Arithmetic on remainders

$\mathbb{Z}_m = \{0, 1, \ldots, m - 1\}$ with + and × done mod m. It is a commutative ring with unity.

IN Z_13

$(17 + 5) \bmod 13 = 9$

$(17 \times 5) \bmod 13 = 7$

WHAT “RING” GIVES YOU

Closure, associativity, commutativity, distributivity, a zero, a one, and negatives for every element.

CAUTION

Division is not always possible in a ring. That is why inverses need a condition. Next lesson block.

QUIZ 3 · 1:08

Questions 1–3

QUESTION 1

a ≡ b (mod m) means…

- **A** a = b
- **B** a + b = m
- **C** m divides (a − b)
- **D** a · b ≡ m

QUESTION 2

Which congruence is true?

- **A** 17 ≡ 5 (mod 7)
- **B** 23 ≡ 3 (mod 5)
- **C** 19 ≡ 4 (mod 6)
- **D** 30 ≡ 9 (mod 8)

QUESTION 3

How many congruence classes does modulus 9 have?

- **A** 9
- **B** 8
- **C** 10
- **D** Infinitely many

Vote by show of hands, then reveal on the next slide.

QUIZ 3 · ANSWERS

Answers 1–3

QUESTION 1

a ≡ b (mod m) means…

Answer: C

m divides (a − b)

Congruent means a and b differ by a multiple of m.

QUESTION 2

Which congruence is true?

Answer: B

23 ≡ 3 (mod 5)

23 − 3 = 20 and 5 | 20. The other differences are 12, 15 and 21.

QUESTION 3

How many congruence classes does modulus 9 have?

Answer: A

9

The classes are [0], [1], …, [8], one per possible remainder.

QUIZ 3 · 1:12

Questions 4–6

QUESTION 4

True or false: Z_m is a commutative ring with unity.

- **A** True
- **B** False

QUESTION 5

What is (17 × 5) mod 13?

- **A** 5
- **B** 9
- **C** 22
- **D** 7

QUESTION 6

What is −3 mod 5?

- **A** −3
- **B** 2
- **C** 3
- **D** 5

Vote by show of hands, then reveal on the next slide.

QUIZ 3 · ANSWERS

Answers 4–6

QUESTION 4

True or false: Z_m is a commutative ring with unity.

Answer: A

True

True. Addition and multiplication mod m behave like a ring, with unity 1.

QUESTION 5

What is (17 × 5) mod 13?

Answer: D

7

17 × 5 = 85 = 6·13 + 7, so the result is 7.

QUESTION 6

What is −3 mod 5?

Answer: B

2

−3 = 5·(−1) + 2, so the remainder is 2.

When can you divide?

a has an inverse mod m if and only if gcd(a, m) = 1

a (mod 12)	gcd	Inverse?	Check
5	1	5	5·5 = 25
7	1	7	7·7 = 49
11	1	11	11·11 = 121
8	4	none	—

EXAMPLE

```
17^(-1) mod 13 = 10
```

Why 8 fails: 8·x is always even, and so is every multiple of 12, so 8x never equals 1 + 12k.

HANDS-ON · 1:22

Use the course-site calculator

INPUT: a = 17, b = 5, m = 13

```
(a + b) mod m = 9
(a × b) mod m = 7
a^b mod m = 10
a^(-1) mod m = 10
```

EUCLID STEPS FOR THE INVERSE

```
13 = 3 × 4 + 1
4 = 4 × 1 + 0
gcd = 1, so it exists.
```

YOUR TURN · 4 MIN

Set a = 7, b = 3, m = 11. Predict all four results on paper first, then check them.

Compute g^e mod m without huge numbers

NAIVE

Compute g^e in full, then reduce. For a 2048-bit exponent the intermediate number is astronomically large.
Not feasible.

SQUARE-AND-MULTIPLY

Use the binary digits of e. Square repeatedly, multiply in when a bit is 1, and reduce mod m after every step.

Cost: O(log e) multiplications instead of e.

PREDICT · 1 MIN

256 = 2^8. How many squarings does 7^256 need?

CODE · 1:30

Square-and-multiply in Python

```
def mod_exp(g, e, m):
    result = 1
    g = g % m
    while e > 0:
        if e % 2 == 1:
            result = (result * g) % m
        g = (g * g) % m
        e = e // 2
    return result
```

TRY IT

```
mod_exp(7, 256, 13)
9
```

IN PRACTICE

Python's built-in pow(g, e, m) does the same job.

WORKED EXAMPLE · 1:35

7^256 mod 13 by repeated squaring

k	7^(2^k)	mod 13
0	7^1	7
1	7^2 = 49	10
2	10^2 = 100	9
3	9^2 = 81	3
4	3^2 = 9	9
5	9^2 = 81	3
6	3^2 = 9	9
7	9^2 = 81	3
8	3^2 = 9	9

7^256 mod 13 = 9

Eight squarings, each on a number below 13. Never a number bigger than 13 × 13.

SPOT THE PATTERN

From k = 3 the values alternate 3, 9. Why?

QUIZ 4 · 1:40

Questions 1–3

QUESTION 1

An inverse of a modulo m exists if and only if…

- **A** a divides m
- **B** gcd(a, m) = 1
- **C** a < m
- **D** m is prime

QUESTION 2

Which number has no inverse modulo 12?

- **A** 5
- **B** 7
- **C** 11
- **D** 8

QUESTION 3

What is the inverse of 17 modulo 13?

- **A** 3
- **B** 4
- **C** 10
- **D** 12

Vote by show of hands, then reveal on the next slide.

QUIZ 4 · ANSWERS

Answers 1–3

QUESTION 1

An inverse of a modulo m exists if and only if…

Answer: B

gcd(a, m) = 1

Coprime is exactly the condition. m prime is sufficient, not necessary.

QUESTION 2

Which number has no inverse modulo 12?

Answer: D

8

gcd(8, 12) = 4, not 1. Inverses: 5·5, 7·7 and 11·11 are all ≡ 1.

QUESTION 3

What is the inverse of 17 modulo 13?

Answer: C

10

17 ≡ 4 and 4 · 10 = 40 = 3·13 + 1, so the inverse is 10.

QUIZ 4 · 1:45

Questions 4–6

QUESTION 4

True or false: Computing g^e in full, then reducing mod m, is practical for 2048-bit numbers.

- **A** True
- **B** False

QUESTION 5

Square-and-multiply computes g^e mod m using…

- **A** O(e) multiplications
- **B** $O(e^2)$ multiplications
- **C** O(√e) multiplications
- **D** O(log e) multiplications

QUESTION 6

What is 7^256 mod 13?

- **A** 9
- **B** 3
- **C** 10
- **D** 7

Vote by show of hands, then reveal on the next slide.

QUIZ 4 · ANSWERS

Answers 4–6

QUESTION 4

True or false: Computing g^e in full, then reducing mod m, is practical for 2048-bit numbers.

Answer: B

False

False. The intermediate number has an astronomically large number of digits.

QUESTION 5

Square-and-multiply computes g^e mod m using…

Answer: D

O(log e) multiplications

One squaring per bit of e, plus at most one multiply per set bit.

QUESTION 6

What is 7^256 mod 13?

Answer: A

9

Squaring chain: 7, 10, 9, 3, 9, 3, 9, 3, 9. After 8 squarings the result is 9.

RECAP · 1:50

Four ideas to keep

Euclid: gcd(a, b) = gcd(b, a mod b), in O(log) steps.

Bézout: a·x + b·y = gcd(a, b), found by Extended Euclid.

Inverses: a^(-1) mod m exists exactly when gcd(a, m) = 1.

Fast powers: square-and-multiply gives g^e mod m in O(log e).

EXIT TICKET · 5 MIN · WRITE ON PAPER

1. Does 6 have an inverse modulo 15? Justify with a gcd.
2. Verify that 23 is the inverse of 7 modulo 40.
3. Reduce −17 modulo 5.